



FACULTÉ DES SCIENCES ET DES TECHNOLOGIES(FST)

TD N°8_ – RÉSEAUX

COURS: RÉSEAUX

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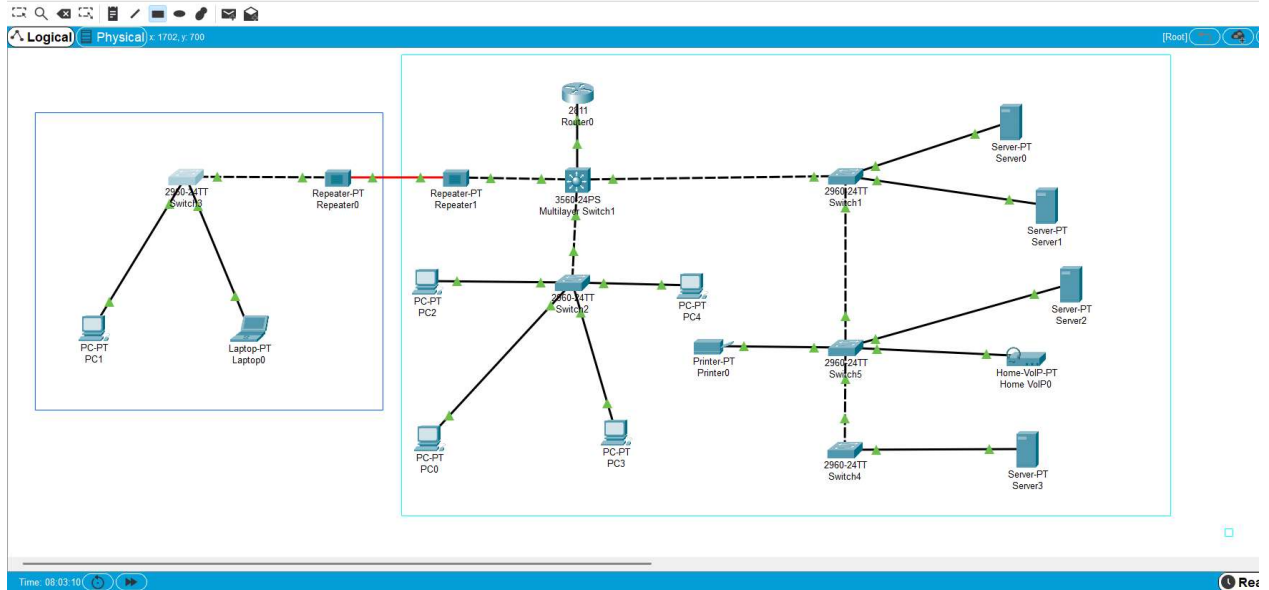
NOM: PÉTIOTE

NIVEAU: L3

DATE: 04/01/2026

L'objectif de ce TD est de:

1. Créer et gérer des VLAN sur des switches
 2. Comprendre le rôle des VLAN pour segmenter un réseau.
 3. Apprendre à créer des VLAN et à affecter des ports aux VLAN correspondants.
 4. Configurer le routage inter-VLAN sur un routeur (Router-on-a-Stick)
 5. Comprendre comment un routeur peut permettre aux VLAN de communiquer entre eux.
 6. Configurer des sous-interfaces avec encapsulation 802.1Q.
 7. Configurer les trunks entre switches et vers le routeur
 8. Apprendre à utiliser les trunks pour transporter plusieurs VLAN sur une seule liaison physique.
 9. Comprendre le concept de VLAN natif et VLAN autorisés sur les trunks.
 10. Attribuer des adresses IP et des gateways aux hôtes
 11. Configurer les PCs dans différents VLAN avec des IP correctes.
 12. Vérifier la communication intra-VLAN (même VLAN) et inter-VLAN (différents VLAN).
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1. Reproduisez cette topologie en créant et configurant plusieurs VLAN sur les switchs. Assignez les ports aux VLAN correspondants, puis testez et gérez la connectivité entre eux à l'aide d'un routeur.



Assignons les ports aux VLAN correspondants

```

sw-campus1>enable
sw-campus1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
sw-campus1(config)#interface f0/1
sw-campus1(config-if)#switchport mode trunk

sw-campus1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
exit
sw-campus1(config)#interface range f0/2-7
sw-campus1(config-if-range)#switchport mode access
sw-campus1(config-if-range)#switchport access vlan 10
sw-campus1(config-if-range)#exit
sw-campus1(config)#interface range f0/
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/3 (10), with swp-campus2 Faexit
sw-campus1#
%SYS-5-CONFIG_I: Configured from console by console

sw-campus1#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/3 (10), with swp-campus2 FastEthernet0/1 (1).
enable
sw-campus1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
sw-campus1(config)#interface f0/1
sw-campus1(config-if)#switchport mode trunk
sw-campus1(config-if)#exit
sw-campus1(config)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/3 (10), with swp-campus2 FastEthernet0/1 (1).
interface range f0/2-6
sw-campus1(config-if-range)#switchport mode access
sw-campus1(config-if-range)#switchport access vlan 10
sw-campus1(config-if-range)#e
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/3 (10), with swp-campus2 FastEthernet0/1 (1).
exit

% Invalid input detected at '^' marker.

sw-campus1(config-if-range)#switchport
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/3 (10), with swp-campus2 FastEthernet0/1 (1).
access vlan 10
sw-campus1(config-if-range)#exit
sw-campus1(config)#interface range f0/8-15
sw-campus1(config-if-range)#switchport mo
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/3 (10), with swp-campus2 FastEthernet0/1 (1).
de access
sw-campus1(config-if-range)#switchport access vlan 20

```

Ctrl+F6 to exit CLI focus

```

Would you like to enter the initial configuration dialog? [yes/no]: enable
% Please answer 'yes' or 'no'.
Would you like to enter the initial configuration dialog? [yes/no]: conf t
% Please answer 'yes' or 'no'.
Would you like to enter the initial configuration dialog? [yes/no]: n

Press RETURN to get started!

Switch>hostname swp-campus2
      ^
% Invalid input detected at '^' marker.

Switch>enable
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname swp-campus2
swp-campus2(config)#vlan 10
swp-campus2(config-vlan)#name ventes
swp-campus2(config-vlan)#vlan 20
swp-campus2(config-vlan)#name marketing
swp-campus2(config-vlan)#vlan 30
swp-campus2(config-vlan)#name it
swp-campus2(config-vlan)#exit
swp-campus2(config)#interface f 0/1
swp-campus2(config-if)#switch mode access
swp-campus2(config-if)#switch mode trunk

swp-campus2(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
interface f 0/2
swp-campus2(config-if)#switch mode access
swp-campus2(config-if)#interface f0/1
swp-campus2(config-if)#switchport mode access
swp-campus2(config-if)#switchport mode trunk

swp-campus2(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
interface f0/2
swp-campus2(config-if)#switchport mode access
swp-campus2(config-if)#switchport mode trunk
swp-campus2(config-if)#exit
swp-campus2(config)#interface f0/24
swp-campus2(config-if)#switchport mode access
swp-campus2(config-if)#switchport mode trunk

```

Ctrl+F6 to exit CLI focus

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swp-campus2>enable
swp-campus2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
swp-campus2(config)#vtp domain monentreprise.com
Changing VTP domain name from NULL to monentreprise.com
swp-campus2(config)#v
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with sw-campus1 FastEthernet0/3 (10).
tp password 123456
Setting device VLAN database password to 123456
swp-campus2(config)#vtp mode server
Device mode already VTP SERVER.
swp-campus2(config)#vtp version 2
swp-campus2(config)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with sw-campus1 FastEthernet0/3 (10).
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with sw-campus1 FastEthernet0/3 (10).

```

```

tp password 123456
Setting device VLAN database password to 123456
swp-campus2(config)#vtp mode server
Device mode already VTP SERVER.
swp-campus2(config)#vtp version 2
swp-campus2(config)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with sw-campus1 FastEthernet0/3 (10).
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with sw-campus1 FastEthernet0/3 (10).
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with sw-campus1 FastEthernet0/3 (10).
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with sw-campus1 FastEthernet0/3 (10).

swp-campus2(config)#spanning-tree mode rapid-pvst
swp-campus2(config)#spanning-tree vlan 1
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with sw-campus1 FastEthernet0/3 (10).
0 root primary
swp-campus2(config)#spanning-tree vlan 20 root secondary
swp-campus2(config)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with sw-campus1 FastEthernet0/3 (10).
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with sw-campus1 FastEthernet0/3 (10).
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with sw-campus1 FastEthernet0/3 (10).
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with sw-campus1 FastEthernet0/3 (10).

sw-campus3>
sw-campus3>enable
sw-campus3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
sw-campus3(config)#interface f0/1
sw-campus3(config-if)#switchport mode trunk

sw-campus3(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
exit
sw-campus3(config)#interface range f0/2-8
sw-campus3(config-if-range)#switchport mode access
sw-campus3(config-if-range)#switchport access vlan 10
sw-campus3(config-if-range)#exit
sw-campus3(config)#interface range f
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/4 (10), with Switch FastEthernet0/5 (
0/8-15
sw-campus3(config-if-range)#switchport mode access
sw-campus3(config-if-range)#switchport access vlan 20
sw-campus3(config-if-range)#exit
sw-campus3(config)#interface range f
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/4 (10), with Switch FastEthernet0/5 (

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Ctrl+F6 to exit CLI focus

Press RETURN to get started!

```
Router>enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface f0/0.10
Router(config-subif)#encapsulation dotQ 10
^
% Invalid input detected at '^' marker.

Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.10.1 155.255.255.0
Bad mask 0x9BFFFF00 for address 192.168.10.1
Router(config-subif)#exit
Router(config)#interface f0/0.20
Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#exit
Router(config)#interface f0/0.30
Router(config-subif)#encapsulation dot1Q 30
Router(config-subif)#ip address 192.168.30.1 255.255.255.0
Router(config-subif)#exit
Router(config)#interface f0/0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/0.10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.10, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/0.20, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.20, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/0.30, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.30, changed state to up
exit
Router(config)#
```

Ctrl+F6 to exit CLI focus

```
swp-campus2 (config)#spanning-tree rapid-pvst
^
% Invalid input detected at '^' marker.

swp-campus2 (config)#spanning-tree mode rapid-pvst
swp-campus2 (config)#spanning-tree vlan 10 root primary
swp-campus2 (config)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on
FastEthernet0/1 (1), with sw-campus1 FastEthernet0/3 (10).
spanning-tree vlan 20 root secondary
swp-campus2 (config)#interface f0/1
swp-campus2 (config-if)#switchport port-security
swp-campus2 (config-if)#switchport port-security
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on
FastEthernet0/1 (1), with sw-campus1 switchport port-security
maximum 3
swp-campus2 (config-if)#switchport port-security violation
restrict
swp-campus2 (config-if)#exit
swp-campus2 (config)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on
FastEthernet0/1 (1), with sw-campus1 FastEthernet0/3 (10).
```

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Vérification important

```
swp-campus2#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/3, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1, Gig0/2
10	ventes	active	
20	marketing	active	
30	it	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
swp-campus2#  
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with sw-campus1 FastEthernet0/3 (10).
```

Ctrl+F6 to exit CLI focus

Test de connectivité sur le routeur

```
Router#ping 192.168.10.2  
  
Type escape sequence to abort.  
Sending 5, 100-byte ICMP Echos to 192.168.10.2, timeout is 2  
seconds:  
.....  
Success rate is 0 percent (0/5)  
  
Router#ping 192.168.20.2  
  
Type escape sequence to abort.  
Sending 5, 100-byte ICMP Echos to 192.168.20.2, timeout is 2  
seconds:  
...|
```

Ctrl+F6 to exit CLI focus

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Coclusion.

Ce travail m'a permis de savoir comment créer et gérer des VLAN sur des switches, comprendre le rôle des VLAN pour segmenter un réseau, comprendre le rôle des VLAN pour segmenter un réseau. Donc la tâche est réussie.